BETOBULLET

4 HP · EURORACK FORMAT
FOR VCV RACK 2



BUCKSAW

STEREO SUPER-OSCILLATOR

USER MANUAL · v2.0.0 · October 2026

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Bucksaw squeezes a 21-wave stereo swarm into 4 HP: three voices, each a stack of seven detuned waves with its own pitch input, and one knob that bends every wave from saw to square. Think walls of trance pad, hollow square basses and leads with teeth.

1. ABOUT BUCKSAW

Stack a handful of sawtooth waves, pull them slightly out of tune and the result is the super saw: a dense, shimmering sound that has driven dance music for decades and gives any filter plenty to chew on. Bucksaw takes that idea to the modular: **3 voices × 7 waves = 21 waves**, spread across a stereo field, with a morph that turns every saw into a square.

Bucksaw recreates the published behaviour of the Acid Rain Technology Chainsaw (manual for firmware 1.1). The original firmware is not public, so the detune curve follows the measured Roland JP-8000 super saw that inspired the design; everything else follows the manual.

FEATURES

- Three independent voices, each activated by patching its 1 V/oct input.
- Seven waves per voice: three detuned up, three down and one always at the root.
- Detune and morph controls, each with CV input (−5 V to +5 V).
- Linear FM of all voices.
- Encoder for the root note: semitone steps, or 1-cent fine tune.
- Stereo output; either output alone works as a full mono signal.

INSTALLATION

Copy `Bucksaw-2.0.0-win-x64.vcvplugin` to `Documents/Rack2/plugins` (or `%LOCALAPPDATA%\Rack2\plugins-win-x64`) and restart VCV Rack. The module appears in the browser under the brand **betobullet**.

2. PANEL OVERVIEW



1. **Root encoder**
   Sets the root pitch of all three voices. Each step is one semitone while the NOTE light is on.
2. **NOTE light**
   Encoder in semitone steps.
3. **ALT light**
   Encoder in 1-cent steps (fine tune).
4. **DETUNE**
   Spreads the seven waves of each voice apart along a non-linear curve. Fully counter-clockwise: perfect unison.
5. **MORPH**
   Morphs every wave from sawtooth to square.
6. **DETUNE CV**
   −5 V to +5 V, added to the DETUNE knob.
7. **MORPH CV**
   −5 V to +5 V, added to the MORPH knob.
8. **V/O 1**
   1 V/oct input of voice 1. Patching a cable activates the voice.
9. **FM**
   Linear frequency modulation of all voices, −5 V to +5 V.
10. **V/O 2**
    1 V/oct input of voice 2.
11. **OUT L**
    Left output.
12. **V/O 3**
    1 V/oct input of voice 3.
13. **OUT R**
    Right output.

3. OPERATION

ROOT FREQUENCY

One encoder **(1)** tunes all three voices together. With the **NOTE** light **(2)** on, every detent moves the root by a semitone; click once and the **ALT** light **(3)** takes over, now one cent per detent. Hold the encoder for two seconds and the root jumps back to **C1 (32.7 Hz)**.

> **MOUSE** Drag up/down or use the scroll wheel to turn the encoder; click to toggle NOTE/ALT; hold still for two seconds to reset. Hovering the encoder shows the note name, cents and frequency.

VOICES

A voice wakes up when a cable lands in its pitch input **(8)**, **(10)** or **(12)**; with all three patched you hear 21 waves. The inputs follow 1 V/octave. A voice with no cable is silent: with nothing patched, Bucksaw makes no sound.

DETUNE

DETUNE **(4)** fans out the seven waves of each voice, one fixed on the root, three going sharp and three going flat. Fully down, all seven sit exactly on the same pitch; turning it spreads them apart along a curve that starts gently and opens up dramatically near the end. CV at input **(6)**, from −5 V to +5 V, is added to the knob position (10 V peak-to-peak sweeps the whole range).

MORPH

The MORPH knob **(5)** morphs all waveforms from sawtooth to square, continuously, by mixing each saw with an inverted copy half a period later. CV at input **(7)** is added to the knob position.

FM

CV at input **(9)**, from −5 V to +5 V, controls the linear frequency modulation of all voices: ±5 V swings the frequency from 0 to twice the pitch. The input responds predictably to audio-rate signals.

STEREO OUTPUTS

Within every voice the seven waves are panned across the field, alternating between left **(11)** and right **(13)**. Use both for width, or just one of them: each side still carries the whole sound.

4. PATCH IDEAS

FIRST SOUND

1. Patch the pitch output of a sequencer or MIDI-CV into **V/O 1**.
2. Patch **OUT L** and **OUT R** into a stereo mixer.
3. Turn the encoder up a few octaves from C1, then open DETUNE until the ”chainsaw” appears.

THREE-NOTE CHORDS

Feed three pitches from a chord generator (for example the chord outputs of HARMONION) into V/O 1–3. With DETUNE around noon you get 21 waves of trance pad.

HOLLOW BASS

One voice, MORPH fully clockwise, DETUNE low, into a low-pass filter with an envelope on cutoff.

MOVING TEXTURE

A slow triangle LFO into DETUNE CV and an inverted copy into MORPH CV make the stack breathe.

5. SPECIFICATIONS

ITEM	VALUE
Width	4 HP
Voices	3, each activated by its V/O cable
Waves per voice	7 (3 up, 3 down, 1 at root)
Root	C1 (32.7 Hz) and up, semitone or 1-cent steps
Pitch inputs	1 V/octave
CV inputs	DETUNE, MORPH: −5 V…+5 V added to the knobs
FM	linear, ±5 V = 0…2× frequency
Outputs	stereo L/R, about 10 Vpp per voice, soft limited at ±12 V
Themes	Black, White (context menu)

CREDITS

Design, DSP and panel: betobullet. Behaviour modelled on the Acid Rain Technology Chainsaw user manual (firmware 1.1); Chainsaw is a trademark of its owner and this is an independent recreation. Detune curve after Adam Szabo's analysis of the Roland JP-8000 super saw. Fonts: Black Ops One, Space Mono, Space Grotesk (SIL OFL).